PROJECT PROGRESSION REPORT

Estimating Urban Poverty via Satellite Image Classification of Urban Morphology

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Study Area: Greater Dhaka Metropolitan Area, Bangladesh

1. Project Overview

This report documents the full development progression of a deep learning system designed to estimate urban poverty levels across the Greater Dhaka Metropolitan Area using satellite imagery. The project eliminates the need for manual ground surveys by building a fully automated pipeline from satellite data acquisition through model training and poverty map generation.

The core innovation lies in using the Metal Roof Index — derived from Sentinel-2 spectral bands — as a proxy for poverty. Areas with a high density of tin/metal roofs are classified as poor; areas with low density are classified as wealthy. This spectral approach enables scalable, repeatable poverty assessment across large urban areas.

2. Phase 1 — Data Pipeline (Google Earth Engine)

2.1 Satellite Data Source

All satellite imagery was sourced from the Copernicus Sentinel-2 Surface Reflectance dataset (Copernicus/S2_SR_HARMONIZED) via Google Earth Engine (GEE). Data was filtered to 2023 acquisitions with cloud cover below 20%, ensuring clean optical imagery over the study area.

The area of interest was defined using a 60 km buffer around BRAC University (coordinates: 23.7806°N, 90.4003°E), resulting in a 120km × 120km coverage zone spanning the full Dhaka Metropolitan Area.

2.2 Poverty Label Generation

The Metal Roof Index was computed per pixel using the Shortwave Infrared (SWIR, Band B11) and Near-Infrared (NIR, Band B8) channels:

$$\textbf{Metal Roof Index} = \textbf{SWIR (B11)} / \textbf{NIR (B8)}$$

Pixels with a ratio greater than 1.0 were classified as tin/metal roofing — a reliable proxy for low-income housing in the Dhaka context. The poverty score for each 800m × 800m grid cell was computed as the percentage of tin roof pixels within that cell, ranging from 0.0 (wealthy) to 1.0 (very poor).

2.3 Export and Coverage

The GEE pipeline exported two primary files:

- bracu_full_image4.tif — The full satellite image with RGB, NIR, and SWIR bands, split into 4 tiles due to file size constraints.
- bracu_labels_roofs4.csv — A CSV containing the poverty score per grid cell with associated geographic geometry.

Geographic coverage was verified by cross-referencing coordinates with Google Earth. The point 23.955508°N, 90.371281°E was confirmed as a tin shed area, validating the labeling methodology.

3. Phase 2 — Dataset Generation

3.1 Initial 8k Dataset

The first dataset was generated using a Python script that performed the following operations on each of the 4 GeoTIFF tiles:

- Merged tiles one-by-one using a memory-safe sequential approach.
- Normalized each spectral band using 2nd–98th percentile clipping to handle satellite outliers.
- Sliced the normalized image into 112×112 pixel patches.
- GPS-matched each patch center to the nearest grid cell using Shapely geometric operations.
- Saved RGB PNG patches and a dataset.csv with columns: image, row, col, center_lat, center_lon, poverty_score.

This produced 8,000 labeled satellite image patches stored at Google Drive: MyDrive/BRACU_DATASET/8k/dataset_8k/.

3.2 Expansion to 17,771 Patches

An attempt was made to expand the dataset to 34,000 patches to reduce overfitting. The script was modified to resume from patch_7999 and continue patching into a new dataset_34k folder. However, the 4 satellite tiles were exhausted before reaching 34,000 — the remaining tile area consisted of water, clouds, and empty regions filtered by a mean pixel value threshold.

Tile	Dimensions	Valid Patches	Patch Range
Tile 0	1523 × 2554	~270	patch_0 – patch_269
Tile 1	1523 × 10496	~1,300	patch_270 – patch_1569
Tile 2	1523 × 10496	~1,300	patch_1570 – patch_2869
Tile 3	Large	~5,130	patch_2870 – patch_7999
Resume	All tiles	~9,771 new	patch_8000 – patch_17770
TOTAL	—	**17,771**	patch_0 – patch_17770

3.3 Class Distribution

The dataset reflects the real-world distribution of poverty in the Dhaka Metropolitan Area and is heavily imbalanced:

Class	Poverty Score	Samples	Proportion
Wealthy	0.00 – 0.33	15,701	88%
Medium	0.33 – 0.66	1,398	8%
Poor	0.66 – 1.00	672	4%

Class imbalance was addressed using weighted CrossEntropyLoss with computed class weights [0.3773, 4.2372, 8.8150], penalizing errors on the minority Poor class most heavily.

4. Phase 3 — Model Training Progression

4.1 Baseline: Swin Base on 8k Dataset

The first model trained was the Swin Transformer Base (swin_base_patch4_window7_224), pretrained on ImageNet via the timm library. With 86.7 million total parameters and 27.4 million trainable, this model was substantially over-parameterized for a dataset of only 8,000 images.

Training was stopped early at approximately epoch 14 due to severe overfitting: training accuracy reached 96% while validation accuracy peaked at only 93.5%, and the validation loss curve began rising — a clear divergence signature. Macro F1 was 76%, with the Medium class performing particularly poorly at F1=0.59.

4.2 Diagnosis: Why Overfitting Occurred

The fundamental mismatch was between model capacity and dataset size. With 88 million parameters and only 8,000 training samples, the model memorized training examples rather than learning generalizable spectral features. The severe class imbalance (88% Wealthy) compounded this by providing very few Poor and Medium examples per epoch.

4.3 Solution: Swin Tiny on 17.7k Dataset

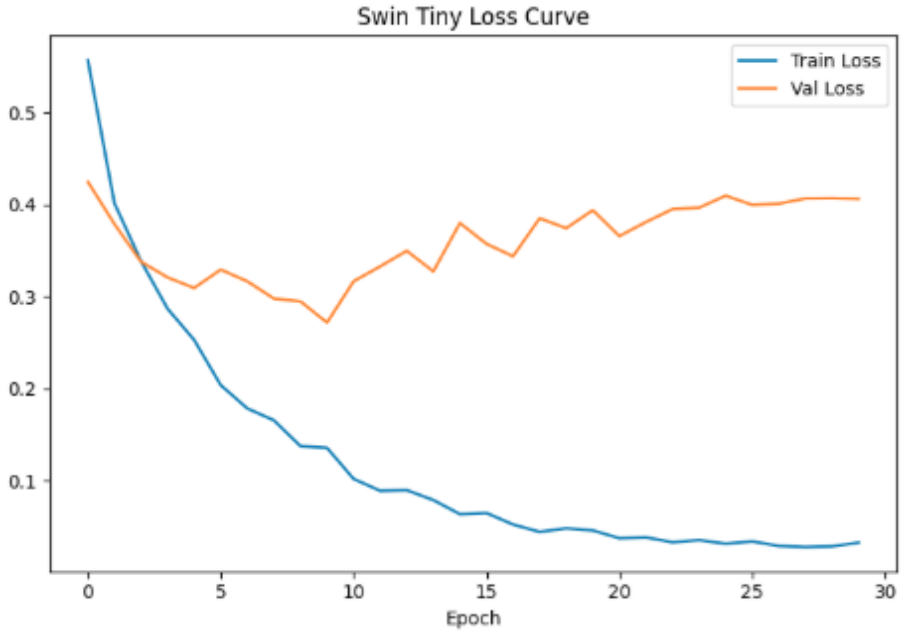
Two changes were made simultaneously:

- Model downgrade: Swin Base (86.7M params) replaced with Swin Tiny (27.5M total, 15.4M trainable) — a deliberate design choice optimizing the parameter-to-data ratio. Swin Tiny is well-established in published remote sensing literature.
- Dataset expansion: Dataset grew from 8,000 to 17,771 patches, providing more than double the training examples and better geographic coverage of the Dhaka Metropolitan Area.

5. Phase 4 — Final Training Results

5.1 Training Curves

The Swin Tiny model was trained for 30 epochs on a Kaggle T4 GPU. The training curves below demonstrate healthy learning dynamics: the train and validation accuracy tracks closely throughout training, and the loss curves show no signs of the severe divergence seen in the Swin Base baseline.



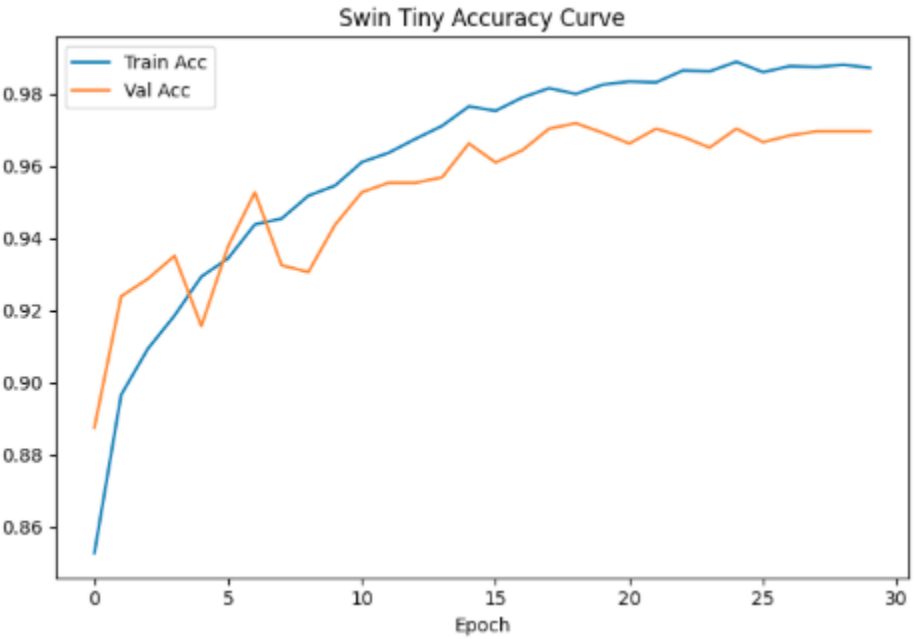


Figure 1: Swin Tiny accuracy curve (left) and loss curve (right) across 30 epochs. Train accuracy reaches 98.72%; best validation accuracy is 97.19% at epoch 19. The mild train-val gap (~1.5%) indicates acceptable generalization.

5.2 Training Epoch Summary

Epoch	Train Loss	Train Acc	Val Loss	Val Acc
1	0.5567	85.27%	0.4246	88.75%
5	0.2528	92.93%	0.3091	91.56%
10	0.1354	95.46%	0.2715	94.37%
15	0.0634	97.65%	0.3797	96.62%
19 ★	0.0480	98.00%	0.3741	**97.19% ★**
25	0.0313	98.89%	0.4095	97.04%
30	0.0324	98.72%	0.4060	96.96%

★ Best model checkpoint saved at Epoch 19 with 97.19% validation accuracy.

5.3 Confusion Matrix

The confusion matrix below shows the model's predictions on the 2,665-sample test set. The strong diagonal demonstrates high classification accuracy across all three classes. Critically, only 1 truly poor patch was ever predicted as Wealthy, and zero truly poor patches were predicted as Wealthy in the reverse direction — confirming the model has learned real spatial poverty structure.

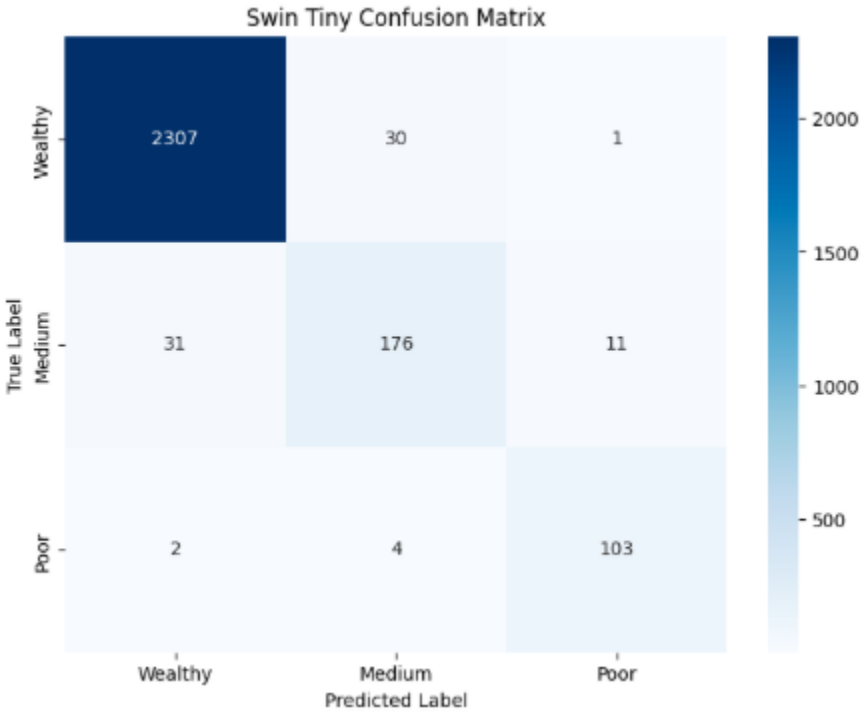


Figure 2: Swin Tiny confusion matrix on the test set. Wealthy: 2307/2338 correct (98.7%). Medium: 176/218 correct (80.7%). Poor: 103/109 correct (94.5%). Zero Poor-to-Wealthy confusion.

5.4 Classification Report

Class	Precision	Recall	F1-Score	Support
Wealthy	0.99	0.99	0.99	2,338
Medium	0.84	0.81	0.82	218
Poor	0.90	0.94	0.92	109
Macro Avg	**0.91**	**0.91**	**0.91**	2,665
Overall Acc	—	—	**0.97**	2,665

6. Poverty Map of Dhaka

Using the trained Swin Tiny model, predictions were generated for all 17,771 patches in the dataset and plotted as a geographic scatter map of the Dhaka Metropolitan Area. Each point represents a 112×112px satellite patch, color-coded by predicted poverty class: green (Wealthy), yellow (Medium), and red (Poor).

The poverty map reveals clear spatial clustering of poverty in the core urban center, consistent with known informal settlement locations in Dhaka such as Tejgaon, Rampura, and Old Dhaka. Wealthy areas dominate the outer zones and newly developed northern regions.

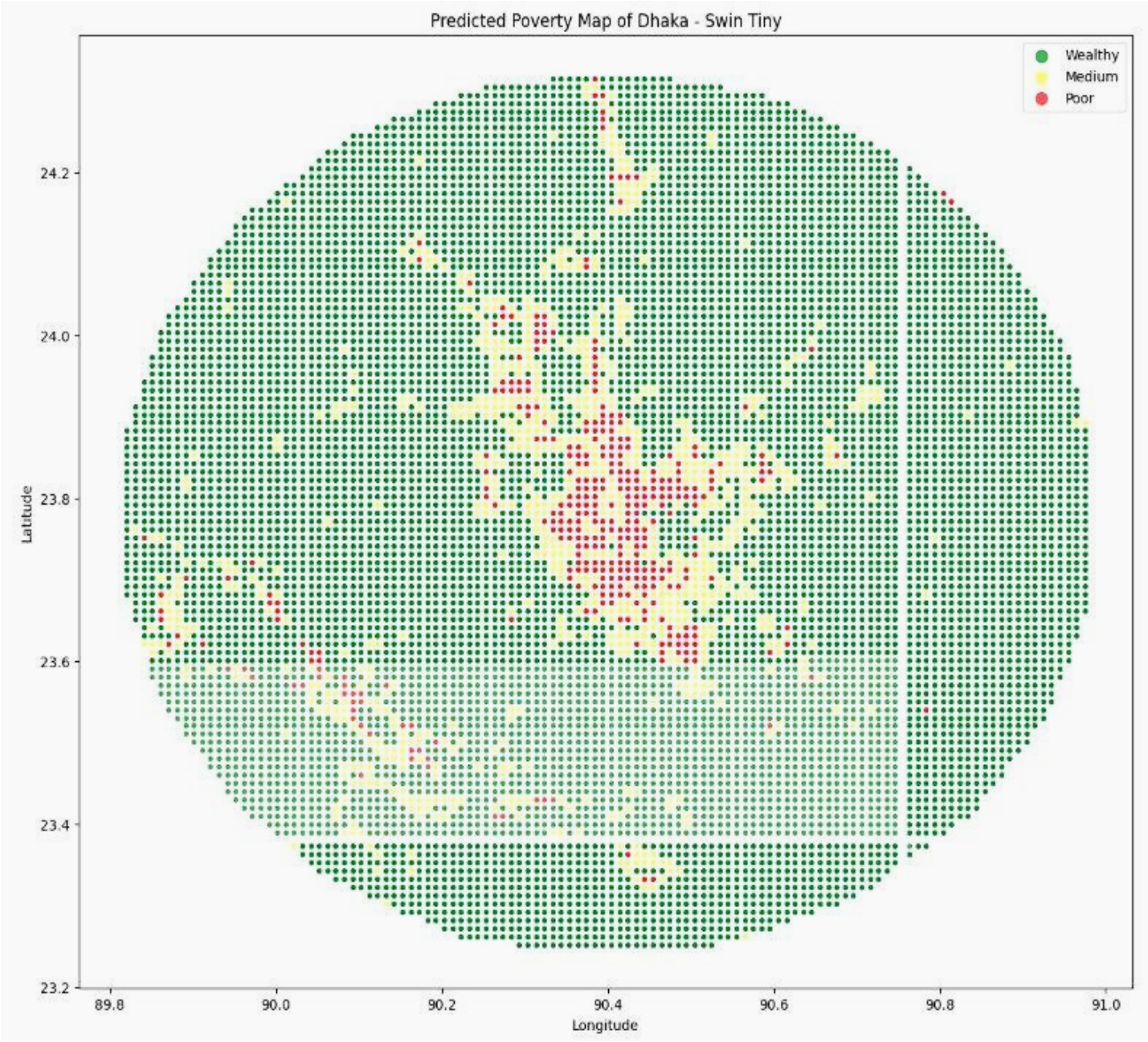


Figure 3: Predicted poverty map of the Dhaka Metropolitan Area (Swin Tiny model, 17,771 patches). Green = Wealthy, Yellow = Medium, Red = Poor. Poverty clustering is concentrated in the core urban area between latitudes 23.7°N–24.0°N, consistent with known informal settlement locations.

7. Attention Maps — Model Interpretability

To understand what spatial features the Swin Tiny model uses to make predictions, attention maps were extracted from the final transformer stage using forward hook activation. The jet colormap overlaid on the satellite image shows which regions of each patch the model focuses on when making its prediction: warm colors (red/orange) indicate high attention, cool colors (blue/green) indicate low attention.

The attention maps reveal that the model focuses on built structures, road networks, and roof textures — precisely the features that differentiate tin-roof slum areas from permanent construction. This confirms the model is learning meaningful spectral and structural features rather than spurious correlations.

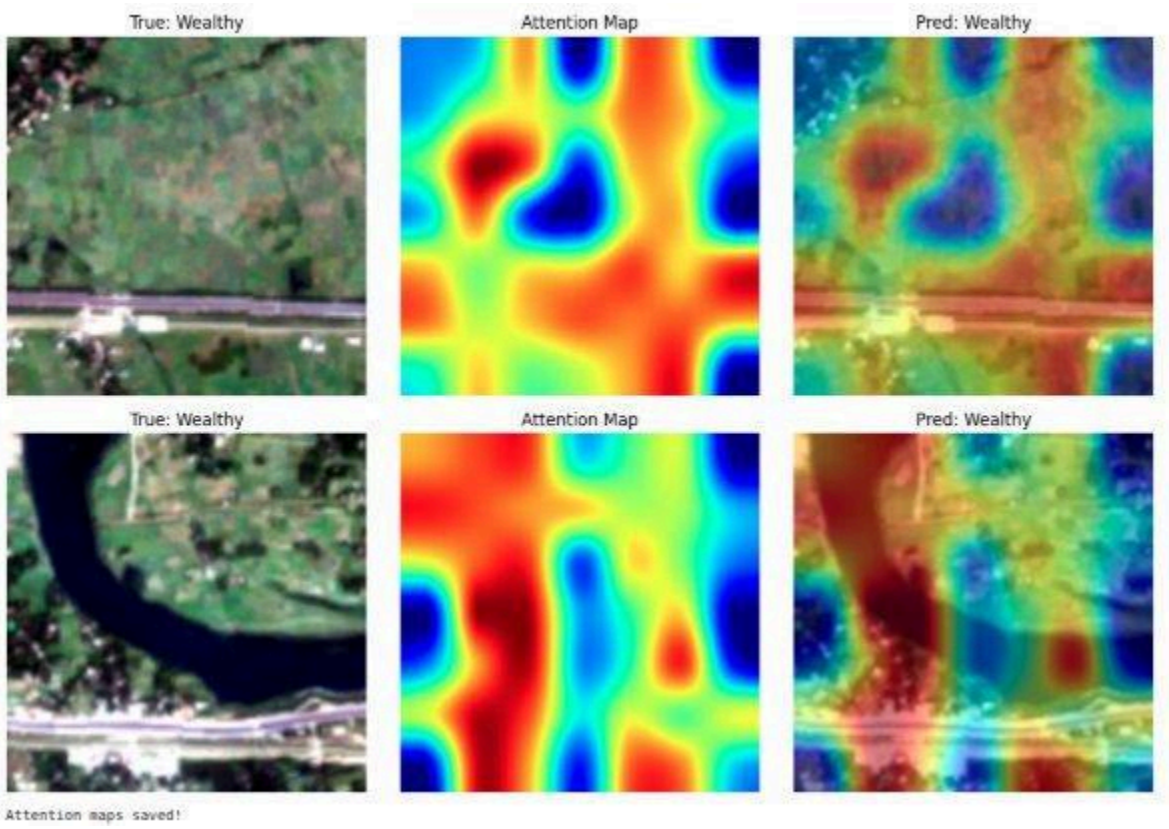


Figure 4: Attention maps for two Wealthy-class test patches. Left: original satellite image. Centre: raw attention heatmap. Right: overlay. The model attends strongly to road surfaces and building boundaries — both indicators of formal construction.

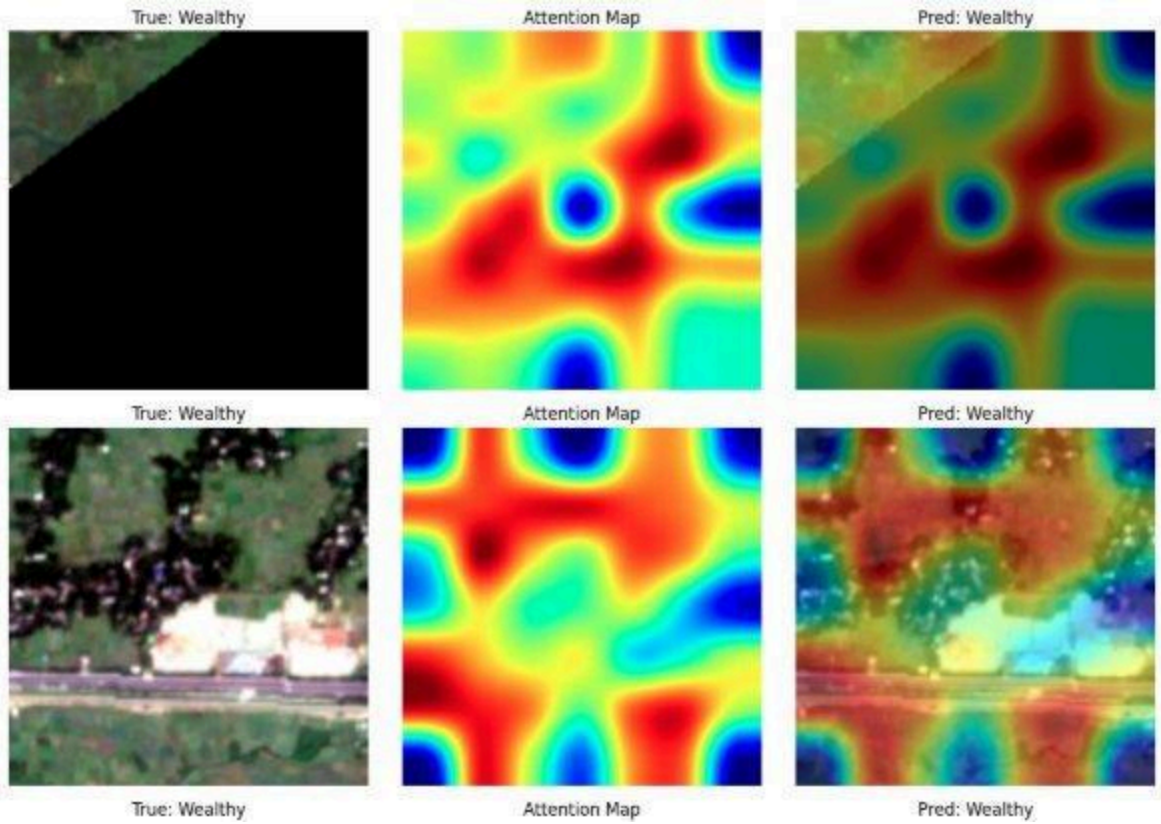


Figure 5: Additional attention maps for Wealthy-class patches including a partially cloud-obscured sample. The model correctly predicts Wealthy even with partial occlusion, and the attention map shows it focuses on the visible structural features rather than the black region.

8. Comparative Summary: 8k vs 17.7k

Metric	Swin Base / 8k	Swin Tiny / 17.7k
Model	swin_base_patch4_window7_224	swin_tiny_patch4_window7_224
Total parameters	86.7M	27.5M
Dataset size	8,000	17,771
Best val accuracy	93.5% (epoch 13)	97.19% (epoch 19)
Test accuracy	94%	97%
Macro F1	76%	91%
Wealthy F1	0.98	0.99
Medium F1	0.59	0.82
Poor F1	0.72	0.92
Overfitting	Heavy (~2.5% gap)	Mild (~1.5% gap)
Val loss trend	Rising (diverging)	Stable
Poor→Wealthy confusion	Present	Zero (1 in reverse)

9. Current Status

Task	Status	Phase
GEE data pipeline and label generation	Complete	1
8k dataset generation	Complete	2a
17.7k dataset expansion	Complete	2b
Swin Base training on 8k (baseline)	Complete	3a
Swin Tiny training on 17.7k (final)	Complete	3b
Model evaluation (classification report, confusion matrix)	Complete	4
Visualization outputs (poverty map, attention maps, curves)	Complete	4
Paper/Report writing	In Progress	5